Database Normalisation

Normalisation organises the columns and tables of a relational database to reduce data redundancy and improve data integrity. A table is in first normal form when every column holds a single atomic value and each row is uniquely identifiable by its key. Repeating groups are moved into a separate related table so that no column stores a list of values.

Second and Third Normal Form

A table is in second normal form when it is in first normal form and every non-key column depends on the whole primary key rather than on part of a composite key. Partial dependencies are removed by splitting the table so each fact is stored exactly once.
Third normal form additionally removes transitive dependencies, where a non-key column depends on another non-key column.

Denormalisation Trade-offs

Denormalisation reintroduces redundancy deliberately to reduce the number of joins a query must perform. The cost is that the same fact is now stored in more than one place, so every write path must keep those copies consistent or the data drifts apart over time.